

SHAMBHU BHANDARI SHARMA

📍 Gower St, London WC1E 6BT @ shambhu.sharma.22@ucl.ac.uk

in [LinkedIn](#)  [GitHub](#)  [Google Scholar](#)

RESEARCH EXPERIENCE

Ph.D Researcher, University College London (2022 Dec–present)

- Developed *ab initio* methods to compute high-pressure phase diagrams of binary alloys.
- Designed and automated computational workflows for melting curve determination.
- Developed and validated classical and machine-learning interatomic potentials.
- Investigated pressure-driven solid–solid phase transitions from first principles.
- Studied premelting effects and mechanical properties of metals and alloys.
- Modelled water-splitting catalysts for hydrogen production, electride materials, and defects in collaboration with experiment.

Research Associate, Khalifa University of Science and Technology (2022 Jan–Nov)

- Performed high-throughput DFT calculations to screen and predict novel 2D materials.
- Investigated facet-dependent properties of crystals for photocatalytic applications.
- Supervised students for the project design and execution.

Master’s Researcher, Tribhuvan University & Kathmandu University (2019–2022)

- Investigated structural, electronic, mechanical, magnetic, and optical properties of 2D materials and nanowires.
- Computed quantum transport properties, including I–V characteristics.
- Utilised national HPC facilities for large-scale simulations.

EDUCATION

University College London (2022–2026)

- PhD in Physics: *ab initio* methods for phase-diagram calculations of binary alloys in the low solute-concentration limit.
- Thesis submitted; viva expected October 2026.

Tribhuvan University (2014–2020)

- MSc in Physics (2020): *ab initio* electronic and quantum-transport properties of gold nanowire.
- BSc in Physics (2014).

TEACHING EXPERIENCE

Teaching Assistant, University College London (2023–2026)

- **Deep Earth & Planetary Modelling (GEOL0046):** molecular dynamics, free-energy methods, and phase boundary calculations on HPC; supervised a Master’s student.
- **AI for Earth Observation (GEOL0069):** transformers, Gaussian processes, and deep learning applied to satellite datasets; deployed ML models on AWS for student outreach.
- **Global Geophysics (GEOL0012):** practicals and coursework marking for 36 students, covering thermodynamics, solid-state physics, and MATLAB programming.

🏆 AWARDS

ISAD Award & EPSRC Industrial CASE Studentship, UCL (2022–present)

- Awarded the International Scholars Award for Doctoral Training (ISAD)
- Recipient of an EPSRC Industrial CASE Studentship

⚙️ SKILLS & EXPERTISE

Computational Condensed-Matter & Materials Physics

- **Electronic Structure (DFT):** Ab initio calculations with VASP, SIESTA, and tranSIESTA, including DFT+U and NEGF methods.
- **Molecular Dynamics & Statistical Mechanics:** Classical and ab initio MD (LAMMPS, AIMD) with free-energy methods, coexistence techniques, and thermodynamic integration.
- **Machine-Learning Interatomic Potentials:** Training and validation using MACE, DeepMD-Kit, and MAML.
- **Atomistic Modelling & Analysis:** Pre- and post-processing with Pymatgen, ASE, OVITO, and VESTA.

Machine Learning & Artificial Intelligence

- **Machine & Deep Learning:** End-to-end predictive models with scikit-learn, Keras, PyTorch, and TensorFlow.
- **Generative AI & LLM Agents:** Agentic systems with RAG pipelines, LoRA fine-tuning, and applications on LangChain and LLM APIs.
- **Data Engineering & Analytics:** ETL/ELT pipelines (Kafka, Airflow), large-scale analysis (Pandas, Spark), dashboards (Plotly, Streamlit), and SQL, NoSQL, and vector databases.

Scientific Computing & Tools

- **Programming:** Python, SQL, FORTRAN, and Bash, with version control via Git/GitHub; development in VS Code and Antigravity IDE.
- **HPC, Cloud & MLOps:** Parallel computing on Linux clusters (SLURM, PBS), Docker containerisation, and AWS resources (SageMaker, Bedrock) via Terraform.
- **Documentation:** Reproducible scientific writing in L^AT_EX and interactive notebooks in Jupyter and Google Colab.

🌟 COURSES & CERTIFICATES

- **Machine Learning Specialization**
By DeepLearning.AI & Stanford (Andrew Ng), completed all 3 courses (≈ 80 hours).
- **Deep Learning Specialization**
By DeepLearning.AI, completed all 5 courses (≈ 120 hours).
- **IBM Data Science Professional Certificate**
By IBM, completed all 11 courses (≈ 160 hours).
- **Data Engineering Professional Certificate**
By DeepLearning.AI & AWS, completed all 4 courses (≈ 120 hours).
- **Generative AI Engineering with LLMs Specialization**
By IBM, completed all 7 courses (≈ 50 hours).

☞ CONFERENCES AND WORKSHOPS

- Regular presenter at national and international conferences; such as TYC PhD Student Conference (2025) and AWE Physics Student Conference, Oxford (2023 & 2025).
- Regular attendee of leading academic and industry summits, including the AE Global Summit on Open Problems for AI (London, 2024) and AWS Summit London (2026).

👤 VOLUNTEER ACADEMIC SERVICE

Peer Reviewer, International Physics & Materials Journals (2022–present)

- Completed 13 invited peer reviews across 8 journals; verified on  [ORCID](#).

📖 PUBLICATIONS

Selected Publications (14 peer-reviewed journal articles and 2 conference proceedings)¹

1. **S. B. Sharma***, *arXiv:2608.29665* **2026** (Evaluating a 4B open-weights local LLM for agentic DFT workflows: a literature reproducibility audit)
2. **S. B. Sharma***, *J. Chem. Inf. Model.* **2026** (Comment on Hybrid Graph-Machine Learning Framework for Accurate and Interpretable Band Gap Prediction, Under Review)
3. **S. B. Sharma***, S. Mehta, and D. Alfè, *arXiv:2608.02225* **2026** (Ab initio High-Pressure Phase Diagrams of Al–Mg Alloys in the Low Solute Concentration Limit)
4. **S. B. Sharma***, S. Mehta, and D. Alfè, *J. Chem. Phys.* **2025**, 162, 184502
5. D. Thapa, **S. B. Sharma**, D. Alfè, D. Kilin, and S. Kilina, *Mater. Horiz.* **2024**, 11, 3613–3622
6. N. Kumar, **S. B. Sharma**, D. Alfè, R. Pandey, and N. Tit, *Appl. Surf. Sci.* **2026**, 744, 167411
7. **S. B. Sharma***, R. Paudel, R. Adhikari, G. C. Kaphle, and D. Paudyal, *Physica E* **2023**, 146, 115517
8. **S. B. Sharma**, I. Qattan, M. Maguy, and A. Sufian, *Comput. Mater. Sci.* **2023**, 218, 111952
9. **S. B. Sharma***, I. Qattan, M. Jaishi, and D. Paudyal, *ACS Appl. Electron. Mater.* **2022**, 4, 2561–2569
10. **S. B. Sharma***, S. KC, and D. Paudyal, *Appl. Surf. Sci.* **2022**, 593, 153239
11. **S. B. Sharma***, and I. Qattan, *Appl. Surf. Sci.* **2022**, 599, 153997

📖 REFERENCES

Prof. Dario Alfè

- University College London, UK.
- d.alfè@ucl.ac.uk · 020 3108 6352

Dr. Shailesh Mehta

- AWE Nuclear Security Technologies, UK.
- s.mehta@awe.co.uk

Dr. Dinesh Thapa

- Thomas More University, USA.
- thapad@thomasmore.edu · 859-344-3367

¹**Bold:** author; * corresponding author. Full list: [Google Scholar](#).